



SAPTHAGIRI NPS
UNIVERSITY

UNMATCHED EXCELLENCE, UNLIMITED POTENTIAL

REPORT ON MINI PROJECT

“SMART TRAFFIC OPTIMIZATION SYSTEM”

Submitted in the partial fulfilment for the award of the degree

**BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE ENGINEERING**

Submitted by

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For the Academic year of 2025-26 [Semester: 4]

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SCHOOL OF ENGINEERING AND TECHNOLOGY

(An ISO 9001:2015 and 14001: 2015 certified Institution)

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CERTIFICATE

Certified that the MINI PROJECT entitled “SMART TRAFFIC OPTIMIZATION SYSTEM” is carried out by AKSHAY N (24SUUBECS0111), AKSHAY GUPTHA.L (24SUUBECS0108), AKSHAY RAVI.B.D (24SUUBECS0113), AKSHAY KRISHNA.H.R(24SUUBECS0109), AKASH.N(24SUUBECS0098) bonafide student of School of Engineering and Technology in partial fulfilment for the award of Bachelor of Engineering in department of Computer Science Engineering of SAPTHAGIRI NPS UNIVERSITY, during the academic year 2025-2026. It is certified that all corrections/suggestions indicated in the Internal Assessment have been incorporated in the report submitted. The mini project report has been approved as it satisfies the academic requirements in respect of mini project prescribed for the Bachelor of Engineering Degree.

Signature of the Guide

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Signature of Director

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ABSTRACT

The **Smart Bengaluru Traffic Optimization System Using Huffman Coding** is a web-based application designed to monitor and optimize traffic conditions in Bengaluru. The system simulates IoT traffic sensors that generate real-time traffic data from different locations such as Rajajinagar, Peenya, Whitefield, and Silk Board. To reduce bandwidth usage and improve data transmission efficiency, the system uses the **Huffman Coding algorithm**, a lossless data compression technique.

The application provides real-time traffic monitoring, compression analytics, congestion detection, and dynamic route recommendations through an interactive dashboard. By combining traffic management with data compression, the system helps improve communication efficiency and supports smarter urban transportation. This project demonstrates the practical application of Huffman Coding, Greedy Algorithms, and real-time web technologies in solving traffic management challenges.

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CHAPTER 1

INTRODUCTION

Transportation plays a vital role in the economic and social development of any city. As urban populations continue to grow, the number of vehicles on roads also increases significantly. This rapid growth often leads to traffic congestion, longer travel times, fuel wastage, environmental pollution, and reduced productivity. Bengaluru, one of India's fastest-growing metropolitan cities and a major information technology hub, faces severe traffic challenges on a daily basis. Managing traffic efficiently has therefore become an important requirement for improving the quality of life of citizens and ensuring smooth transportation.

Traditional traffic management systems mainly rely on manual observation, fixed traffic signals, and periodic monitoring methods. These systems are often unable to respond effectively to rapidly changing traffic conditions. With the emergence of Smart City initiatives and advancements in Internet of Things (IoT) technologies, modern traffic management systems can now collect and analyze real-time traffic information using interconnected sensors and communication networks. Such systems help authorities monitor traffic conditions continuously and make better decisions for traffic control and route planning.

IoT-based traffic sensors can collect large amounts of information from different locations. These sensors generate data such as traffic density, vehicle count, congestion level, average speed, and road conditions. While real-time monitoring provides valuable insights, the continuous generation and transmission of large volumes of data create challenges related to network bandwidth, storage requirements, and processing efficiency. As the number of sensors increases, the amount of data transmitted across the network also grows significantly. Therefore, efficient data management techniques are necessary to ensure smooth and cost-effective communication between sensors and monitoring systems.

Data compression is one of the most effective techniques for reducing the size of transmitted information. Compression techniques minimize the amount of data that needs to be stored or transmitted while preserving the original information. Among various compression methods, Huffman Coding is widely recognized as an efficient lossless data compression algorithm. Huffman Coding is based on the Greedy Algorithm approach and generates variable-length binary codes for

different data values based on their frequency of occurrence. Frequently occurring values receive shorter binary codes, while less frequent values receive longer codes. This results in reduced data size and improved transmission efficiency without any loss of information.

The project titled "**Smart Bengaluru Traffic Optimization System Using Huffman Coding**" aims to integrate real-time traffic monitoring with efficient data compression techniques to create a smart traffic management solution. The system simulates IoT traffic sensors deployed across important locations in Bengaluru such as Rajajinagar, Peenya, Yeshwanthpur, Jalahalli, Hesaraghatta, Whitefield, and Silk Board. These sensors continuously generate traffic-related information including traffic level, vehicle count, congestion percentage, and timestamps. The generated traffic data is then processed and compressed using the Huffman Coding algorithm before being transmitted to the user dashboard.

One of the major advantages of incorporating Huffman Coding into traffic monitoring systems is the reduction of bandwidth consumption. Since traffic data often contains repetitive values such as HIGH, MEDIUM, and LOW traffic levels, Huffman Coding can efficiently compress these repeated patterns. The compressed data requires less storage space and can be transmitted more quickly over communication networks. This improves the overall efficiency of the system and enables real-time monitoring with reduced communication costs.

The proposed system also utilizes modern web technologies to provide an interactive and user-friendly dashboard. The frontend is developed using React and Tailwind CSS, while the backend uses Node.js and Express.js. Real-time communication between the server and client is achieved through Socket.IO, enabling instant updates whenever traffic conditions change.

CHAPTER 2

LITERATURE REVIEW / BACKGROUND STUDY

Traffic congestion has become a major challenge in urban areas due to rapid population growth and increasing numbers of vehicles. Researchers and transportation authorities have developed various intelligent traffic management systems to improve traffic flow, reduce congestion, and enhance road safety. Traditional traffic monitoring systems primarily depend on CCTV cameras, manual traffic surveys, and fixed traffic signal controls. Although these methods provide useful Z

Researchers have also explored data analytics and machine learning techniques for predicting traffic patterns and optimizing transportation systems. These approaches analyze historical and real-time traffic data to forecast congestion levels and recommend alternative routes. Such intelligent systems contribute to reducing travel time, fuel consumption, and environmental pollution. However, one of the major challenges faced by these systems is the large volume of data generated by numerous traffic sensors deployed across different location.

To address the issue of excessive data transmission and storage requirements, data compression techniques have been widely studied. Among various compression algorithms, Huffman Coding is one of the most popular lossless compression methods.

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CHAPTER 3

PROBLEM DEFINITION, OBJECTIVES AND METHODOLOGY

3.1 Problem Definition

Traffic congestion is one of the major challenges faced by rapidly growing cities such as Bengaluru. The increasing number of vehicles on roads leads to delays, fuel wastage, environmental pollution, and reduced transportation efficiency. Modern traffic monitoring systems generate a large volume of real-time sensor data from different locations. Managing, transmitting, and analyzing this data efficiently becomes difficult due to bandwidth limitations and storage requirements.

The main problem addressed in this project is to develop an intelligent traffic monitoring and optimization system that can efficiently collect, compress, and analyze traffic data in real time. Traditional systems often transmit raw sensor data, resulting in increased network traffic and communication overhead. Furthermore, understanding how data compression techniques such as Huffman Coding improve transmission efficiency is difficult without proper visualization.

The proposed Smart Bengaluru Traffic Optimization System Using Huffman Coding provides a real-time traffic monitoring platform that simulates IoT sensors, compresses traffic data using Huffman Coding, and displays the complete compression process through an interactive dashboard. The system helps users understand traffic conditions, data compression techniques, and route optimization while reducing bandwidth consumption and improving communication efficiency.

3.2 Objectives

The primary objective of the Smart Bengaluru Traffic Optimization System Using Huffman Coding is to develop an intelligent and interactive platform for monitoring and optimizing traffic conditions in Bengaluru. The project aims to simulate real-time traffic sensor data and demonstrate how Huffman Coding can be used to efficiently compress traffic information before transmission.

Another objective is to provide a user-friendly dashboard that displays live traffic updates, sensor activity, congestion levels, and compression statistics. The system enables users to visualize traffic conditions at different Bengaluru locations such as Rajajinagar, Peenya, Yeshwanthpur, Jalahalli, Hesaraghatta, Whitefield, and Silk Board.

The project also aims to demonstrate the practical application of Data Structures and Algorithms concepts such as Huffman Trees, Priority Queues, Greedy Algorithms, and Binary Encoding. Through visualization of frequency tables, Huffman Trees, binary codes, and compression ratios, users can better understand the working of Huffman Coding.

Furthermore, the system seeks to provide intelligent route recommendations based on traffic congestion levels. By detecting highly congested areas and suggesting alternate routes, the project helps improve travel efficiency and demonstrates the use of smart traffic management techniques.

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3.3 Methodology

3.3.1 System Design

The system is designed as a full-stack web application that integrates traffic monitoring, Huffman data compression, route optimization, and analytics. The application consists of a React-based frontend, an Express.js backend, Socket.IO for real-time communication, and a Huffman Coding engine for data compression. The dashboard provides a modern interface for visualizing traffic conditions and compression performance.

3.3.2 Traffic Sensor Simulation

The system simulates IoT traffic sensors deployed across multiple Bengaluru locations. These virtual sensors generate traffic data every three seconds, including traffic level, vehicle count, congestion percentage, and timestamp. The generated data represents real-time traffic conditions and serves as input for further processing.

3.3.3 Huffman Coding Implementation

The Huffman Coding algorithm forms the core of the compression module. The system calculates the frequency of traffic values such as HIGH, MEDIUM, and LOW, constructs a Huffman Tree using a Priority Queue (Min Heap), generates binary codes, and compresses the traffic data. The compressed binary stream is then transmitted instead of the original data, reducing bandwidth usage and improving communication efficiency.

3.3.4 Frontend Development

The user interface is developed using React.js, Vite, Tailwind CSS, and Recharts. The frontend provides dashboard pages for traffic monitoring, Huffman visualization, route optimization, and analytics. Interactive cards, charts, animations, and responsive layouts enhance the user experience and make the system visually appealing.

3.3.5 Real-Time Communication and Data Processing

Socket.IO is used to establish real-time communication between the server and the client dashboard. As sensor data is generated and compressed, updates are instantly transmitted to connected users. The dashboard automatically refreshes traffic statistics, congestion alerts, compression metrics, and route recommendations without requiring manual page reloads.

3.3.6 Result Visualization and Analysis

The system provides detailed visualizations of traffic and compression data. Users can view live traffic status, sensor activity, frequency tables, Huffman Trees, binary codes, compression ratios, bandwidth savings, and congestion trends. Recharts are used to generate line charts, bar charts, area charts, and pie charts for effective analysis and interpretation of data.

3.3.7 Testing and Deployment

The application is tested using various traffic scenarios and sensor inputs to ensure accuracy, responsiveness, and reliability. Different traffic levels and congestion conditions are simulated to verify the correctness of compression and route recommendation modules. The project is developed and executed using the Vite development environment and is designed to be deployed on modern web hosting platforms while maintaining scalability and performance.

CHAPTER 4

IMPLEMENTATION -"TRAFFIC OPTIMIZATION SYSTEM "

The Smart Bengaluru Traffic Optimization System Using Huffman Coding is implemented as a full-stack web application that integrates real-time traffic monitoring, data compression, route optimization, and analytics into a single platform. The frontend is developed using React.js, Vite, and Tailwind CSS to create a modern, responsive, and user-friendly dashboard. The backend is built using Node.js and Express.js, which handle data processing, API management, and communication between different modules. Socket.IO is used to establish real-time communication between the server and the client, allowing traffic updates to be transmitted instantly without requiring page refreshes. The system simulates IoT traffic sensors deployed at major Bengaluru locations such as Rajajinagar, Peenya, Yeshwanthpur, Jalahalli, Hesaraghatta, Whitefield, and Silk Board. These virtual sensors generate traffic information every few seconds, including traffic level, vehicle count, congestion percentage, and timestamp.

A key feature of the system is the implementation of the Huffman Coding algorithm, which is used to compress traffic data before transmission. The compression module calculates the frequency of traffic values, constructs a Huffman Tree using a Priority Queue (Min Heap), generates binary codes, and converts traffic messages into compressed binary streams. This process significantly reduces data size, decreases bandwidth consumption, and improves communication efficiency while preserving the original information. The compressed data is transmitted to the frontend dashboard, where users can view the original message, compressed output, frequency table, Huffman tree visualization, and compression ratio. The system also includes a route optimization module that continuously monitors traffic congestion levels and suggests alternate routes whenever a highly congested area is detected. For example, if traffic congestion at Peenya becomes high, the system recommends an alternative route through Jalahalli to reduce travel time.

To enhance data analysis and decision-making, the application provides an advanced analytics dashboard using the Recharts library. Various charts such as line charts, bar charts, area charts, and pie charts are used to visualize traffic density, compression efficiency, bandwidth savings, and congestion trends.

CHAPTER 5

RESULTS AND DISCUSSION

The **Smart Bengaluru Traffic Optimization System Using Huffman Coding** was successfully developed and tested using simulated traffic data generated from multiple Bengaluru locations, including Rajajinagar, Peenya, Yeshwanthpur, Jalahalli, Hesaraghatta, Whitefield, and Silk Board. The system was able to continuously generate real-time traffic information such as traffic levels, vehicle counts, congestion percentages, and timestamps. Through the integration of Socket.IO, the dashboard received live updates without requiring manual refreshes, providing an efficient real-time monitoring experience. The traffic monitoring module accurately displayed current traffic conditions, active sensor status, and congestion alerts, allowing users to easily identify heavily congested areas.

The Huffman Coding module produced successful compression and decompression results for traffic sensor data. Repetitive traffic values such as HIGH, MEDIUM, and LOW were efficiently encoded into shorter binary representations, reducing the amount of data transmitted across the network. The system displayed frequency tables, Huffman trees, binary codes, compressed messages, and compression ratios, helping users understand the compression process. The results showed a noticeable reduction in data size, leading to bandwidth savings and improved transmission efficiency. The route optimization module also performed effectively by identifying congested routes and suggesting alternate paths to minimize travel time. Real-time notifications informed users whenever traffic congestion reached high levels.

The analytics dashboard provided meaningful insights through interactive charts and visualizations. Traffic density trends, compression efficiency, bandwidth savings, sensor activity statistics, and congestion patterns were clearly represented using line charts, bar charts, area charts, and pie charts. The dark-themed responsive dashboard improved usability across different devices and screen sizes.

OUTPUTS:

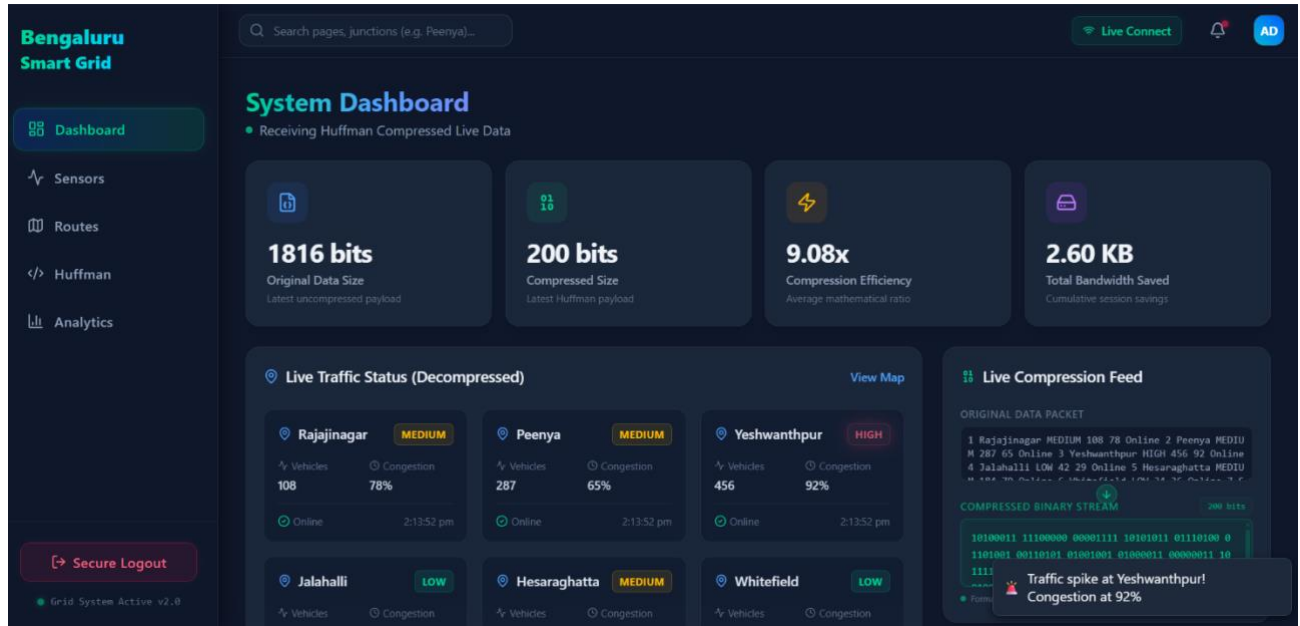


Fig 5.1 – Smart Bengaluru Traffic Optimization System Dashboard

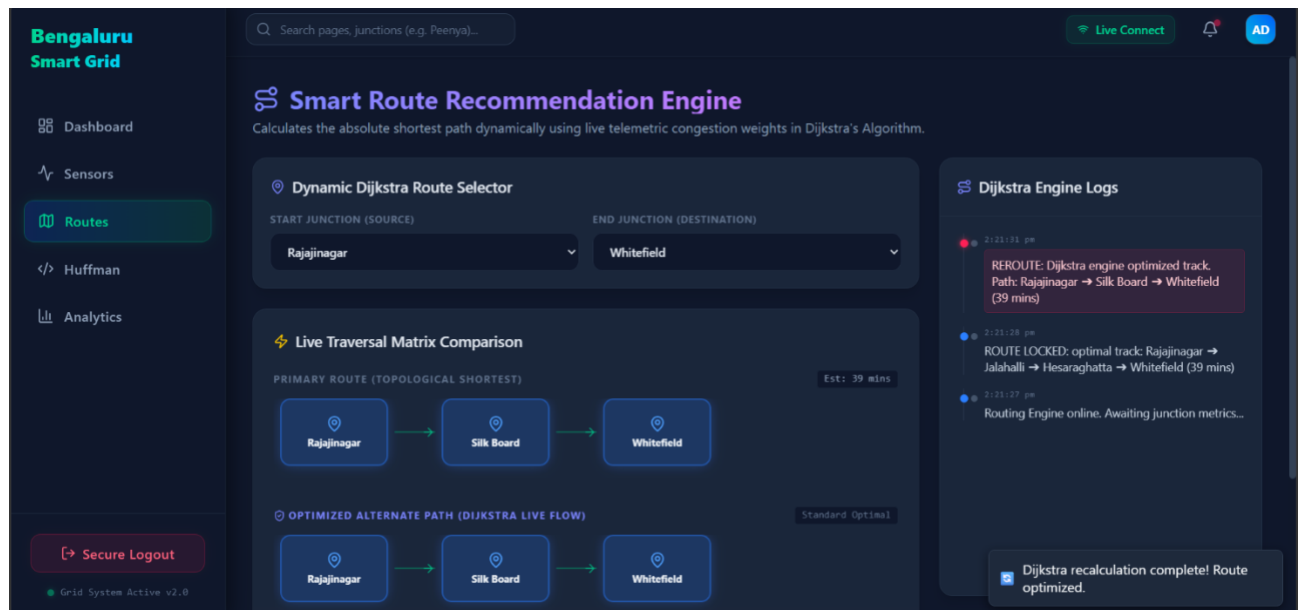


Fig 5.2 – Smart Route Recommendation Engine Using Dijkstra's Algorithm

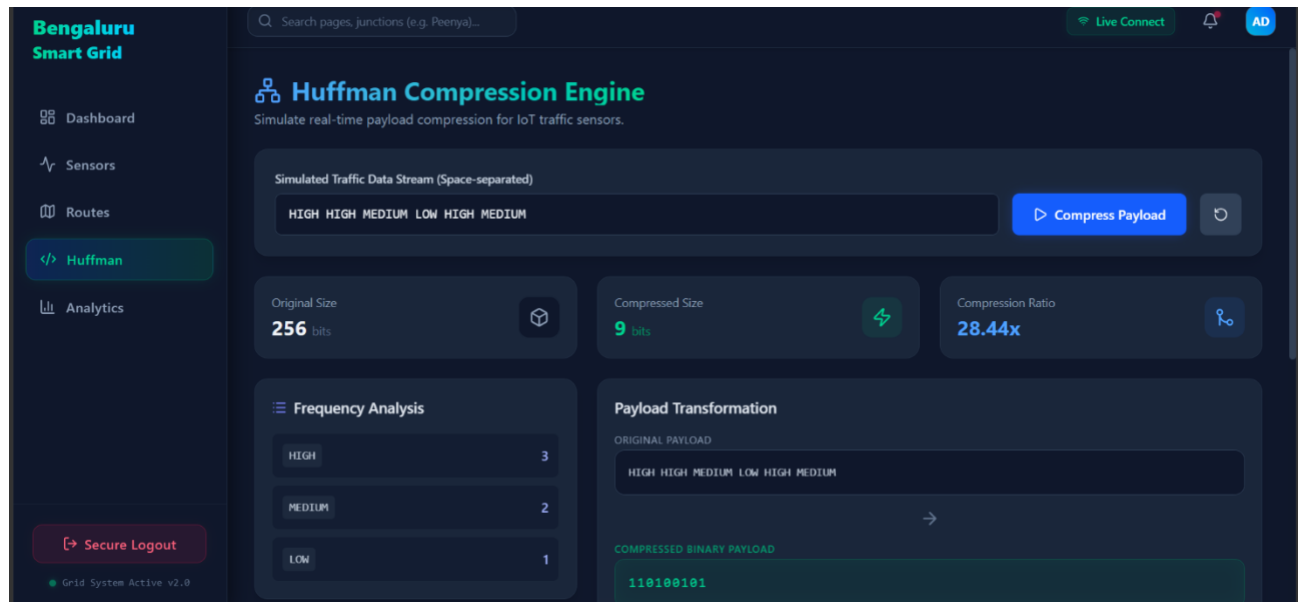


Fig 5.3- Huffman Coding Module for Traffic Data Compression

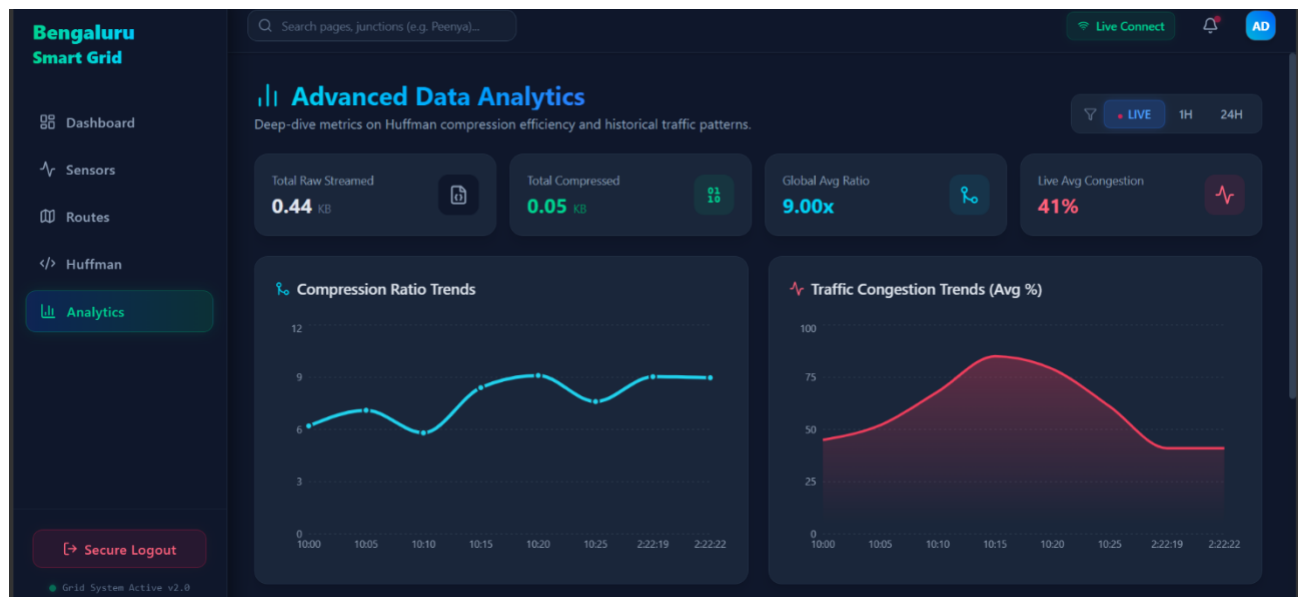


Fig 5.3– Real-Time Traffic Congestion and Compression Analytics

CHAPTER 6

CONCLUSIONS AND SCOPE FOR FUTURE WORK

6.1 Conclusion

The Smart Bengaluru Traffic Optimization System Using Huffman Coding was successfully developed to demonstrate the integration of real-time traffic monitoring, data compression, route optimization, and analytics within a single platform. The system simulates IoT traffic sensors that continuously generate traffic information from various Bengaluru locations and efficiently compresses the data using the Huffman Coding algorithm before transmission. Through an interactive dashboard and real-time visualizations, users can monitor traffic conditions, analyze compression performance, and receive dynamic route recommendations. The implementation of Huffman Coding significantly reduces data size and bandwidth consumption while maintaining data accuracy. The results demonstrate that the system effectively improves communication efficiency, provides meaningful traffic insights, and showcases the practical application of Data Structures and Algorithms concepts in solving real-world traffic management challenges. Overall, the project achieves its objective of developing an intelligent, user-friendly, and educational traffic optimization system suitable for smart city environments.

6.2 Scope for Future Work

The current system focuses on simulated traffic monitoring and Huffman-based data compression. In the future, the project can be enhanced by integrating real IoT traffic sensors and live traffic data sources to provide more accurate traffic analysis. Advanced machine learning and artificial intelligence techniques can be incorporated to predict future traffic congestion and recommend optimal routes proactively. The route optimization module can be extended using advanced graph algorithms and real-time geographic information systems for more precise navigation. Additional features such as accident detection, emergency vehicle prioritization, smart traffic signal control, and integration with smart city infrastructure can further improve system effectiveness.

CHAPTER 7

REFERENCES

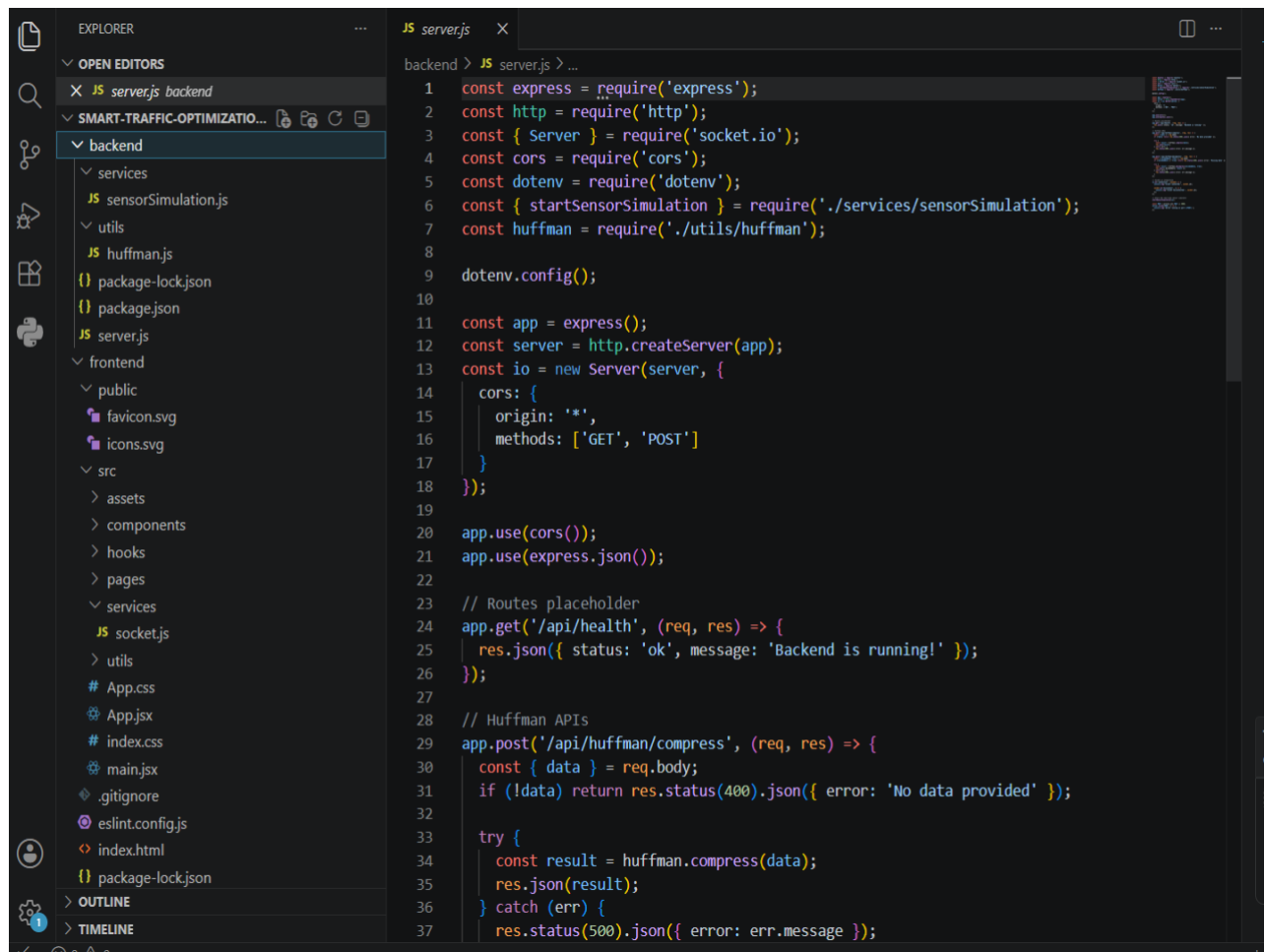
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- MDN Web Docs. JavaScript Guide. Mozilla Developer Network. Available at: <https://developer.mozilla.org/en-US/docs/Web/JavaScript> (Referenced for JavaScript programming concepts and implementat

CHAPTER 8

APPENDIX / ANNEXURE

Sample source code:

8.1 Smart Traffic Optimization System :



The screenshot displays the Visual Studio Code editor interface. On the left, the Explorer sidebar shows the project structure for 'SMART-TRAFFIC-OPTIMIZATION...'. The 'backend' directory is expanded, revealing files like 'sensorSimulation.js', 'utils', 'huffman.js', 'package-lock.json', and 'package.json'. The 'server.js' file is selected and open in the main editor. The code is written in JavaScript and uses ES6 syntax (const, let, arrow functions). It initializes Express.js, sets up CORS, and defines routes for health checks and Huffman compression. The code is as follows:

```
1  const express = require('express');
2  const http = require('http');
3  const { Server } = require('socket.io');
4  const cors = require('cors');
5  const dotenv = require('dotenv');
6  const { startSensorSimulation } = require('./services/sensorSimulation');
7  const huffman = require('./utils/huffman');
8
9  dotenv.config();
10
11 const app = express();
12 const server = http.createServer(app);
13 const io = new Server(server, {
14   cors: {
15     origin: '*',
16     methods: ['GET', 'POST']
17   }
18 });
19
20 app.use(cors());
21 app.use(express.json());
22
23 // Routes placeholder
24 app.get('/api/health', (req, res) => {
25   res.json({ status: 'ok', message: 'Backend is running!' });
26 });
27
28 // Huffman APIs
29 app.post('/api/huffman/compress', (req, res) => {
30   const { data } = req.body;
31   if (!data) return res.status(400).json({ error: 'No data provided' });
32
33   try {
34     const result = huffman.compress(data);
35     res.json(result);
36   } catch (err) {
37     res.status(500).json({ error: err.message });
38   }
39 });
```